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Original paper

Design and validation of an integrated reference dosimetry and monitoring system for ultra-high dose-rate proton beams ranging from 20 Gy/s to 230 Gy/s

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A R T I C L E I N F O

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A B S T R A C T

Purpose: This study aims to design, develop, and test an integrated reference dosimetry and beam monitoring system tailored for proton beams in a dose rate regime ranging from 20 Gy/s to 230 Gy/s. The primary objective is to establish a robust system for diagnostic and dosimetry addressing the challenges associated with ultra-high dose-rate (UHDR) radiotherapy. The proposed detectors chain seeks to ensure accurate dose measurements to support future clinical applications and radiobiological experiments in FLASH proton therapy (pFLASH-RT).

Methods: The system integrates three detectors, a Secondary Emission Monitor (SEM), a Dual-Gap Ionization Chamber (DGIC), and a Faraday Cup (FC) , providing dose measurements. The SEM and DGIC operate continuously to monitor dose rates, while the FC, designed with innovative geometric and electronic features, ensures dose calibration. Experimental validations were conducted using a 62 MeV proton beam at the INFN-LNS, spanning various dose rates. Calibration procedures and correction algorithms, including the Boag–Wilson theory, were applied to ensure the reliability of the dosimetry and monitoring system.

Results: Experimental results demonstrated high reproducibility and accuracy of the entire system. The FC exhibited a mean relative dose uncertainty of 2%, with no significant response variation across dose rates, even at the highest dose rate tested (230 Gy/s). Similarly, the SEM demonstrated consistent performance, with an average agreement within 1.4% of the FC measurements. Additionally, the application of correction factors based on collection efficiency parameters reduced the DGIC measurement uncertainty to less than 3%.

Conclusion: The proposed system represents a reliable solution for the dosimetry of UHDR proton beams. Its ability to provide accurate, real-time dose measurements under extreme beam conditions supports the integration of pFLASH-RT into clinical practice. While the individual detectors employed in this work (Faraday Cup, DGIC, SEM) are based on established technologies, the innovation of this study lies in the integration and cross-calibration of these components within a single real-time dosimetric architecture, experimentally validated over a wide dose rate range. Furthermore, the system's robustness and reproducibility make it an invaluable tool for advancing radiobiological research and ensuring the safe application of UHDR proton therapy.

1. Introduction

The protection of healthy tissues in radiation therapy (RT) has been greatly improved over the last few decades due to enhanced precision in dose delivery, dose conformity, and optimized fractionation schedules. Recent in-vivo and in-vitro studies have shown that RT delivered at ultra-high dose rates (UHDR) regimes, where the dose rates exceed 40 Gy/s, can achieve similar treatment goals while potentially reducing toxicity [1–4]. This technique, known as FLASH-RT, has shown great promise in several studies for improving the efficacy of RT. FLASH-RT is particularly promising for treating cancers where RT is a key

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component of therapy, and where reducing the high rates of acute and late side effects is crucial, such as in adult and paediatric brain tumors [5]

The first patient was successfully treated using FLASH electrons in 2019 for cutaneous lymphoma, showing safe skin tolerance [6]. More recently, a dedicated clinical trial has been launched (https://clinicaltrials.gov/ct2/show/NCT04592887) to investigate the use of proton FLASH-RT (pFLASH-RT) for the treatment of bone metastases. The first results from this trial were reported in the FAST-01 study, which confirmed the feasibility, safety, and palliative efficacy of single-fraction 250 MeV transmission pFLASH-RT in a cohort of patients with extremity bone metastases [7].

Nevertheless, the translation of FLASH-RT into routine clinical practice still depends on the ability to guarantee accurate and real-time dosimetry across a broad dose-rate spectrum Developing and validating reliable beam monitoring systems that maintain performance even under high-dose rate conditions, remains a fundamental challenge to be addressed. Although numerous preclinical studies have been carried out to explore and validate the FLASH effect, the outcomes remain partially compromised due to the absence of standardized methods for dose measurement and beam monitoring. The techniques currently employed in conventional radiotherapy, and adapted for FLASH-RT, depend on detectors that can be significantly affected by the high dose rates and dose per pulse. In proton beam reference dosimetry is generally based on ionization chambers [8]according to the protocols widely accepted in current clinical practice [9] . However they do not provide recommendations for treatments involving ultra-high dose rates. In particular, the lack of well-established real-time in-beam detectors that can accurately monitor dose delivery limits the potential of newly installed facilities, introducing significant uncertainties in the obtained results [10]. Several studies have recently focused on quantifying the performance of ionization chambers under UHDR conditions, particularly in terms of recombination correction and response linearity. In the context of the FAST-01 clinical trial, [11] showed that both the PTW Advanced Markus and the IBA PPC05 parallel-plate chambers, when appropriately corrected using established methods such as the two-voltage technique, could provide dose measurements within 1%–3% agreement with a primary calorimetric reference across dose rates up to 60 Gy, s^{-1}. This indicates that conventional chambers may still serve as reliable dosimeters in proton FLASH settings, provided that appropriate corrections are applied and the beam temporal structure is well characterized. However, passive detectors such as alanine, radiochromic films (RCF), and thermoluminescent dosimeters are currently widely used [12–14]. These detectors require precise calibration and time-consuming post-irradiation processing, and their effectiveness for routine use with pFLASH-RT irradiations has not been thoroughly tested or documented. Furthermore, absolute dosimetry based on primary standard measurements has not yet been fully developed for this emerging modality. Although calorimeters and Faraday Cup (FC) detectors have been successfully utilized, a gold standard detector has not yet been established [14,15]. Calorimetry has recently been demonstrated as a primary standard for absolute dosimetry in proton FLASH radiotherapy, achieving uncertainties below 1%, thus establishing a fundamental benchmark for system calibration under UHDR conditions [16]. Faraday Cups have historically been employed in proton beam dosimetry, as reported by [17–20] and 1979, primarily for low-to-moderate dose rate applications. Their integration into high-intensity proton systems has remained limited, particularly for pulsed beams above 200 Gy/s. Similarly, dual-gap ion chambers have been proposed as dose-rate independent monitors and implemented in commercial systems such as the IBA ProteusOne. Scintillator-based solutions for time-resolved FLASH monitoring have also emerged [22,23], although full integration into proton FLASH workflows remains at an early stage. The integration of the pFLASH-RT modality into clinical practice will require the establishment of robust and accurate dosimetric procedures, as well as the development and characterization of new absolute and relative dosimetric detectors capable of operating effectively under these novel and extreme conditions. In this work, we present an integrated dosimetry and monitoring system composed of three detectors designed able to measure in real-time a dose in terms of Gy in water. The monitoring system is based on the use of two online detectors: a SEM (Secondary Emission Monitor) and a DGIC (Dual-Gap Ionization Chamber). The dose measurement was performed using a Faraday Cup (FC) operating as a fluence detector. The collected charge was converted into absorbed dose in water (Gy) according to the formalism described in [19]. The FC, entirely designed and built at the INFN-LNS (Laboratori Nazionali del Sud – Istituto Nazionale di Fisica Nucleare), was specifically optimized for high-intensity, pulsed proton beams, and employed here for dosimetric purposes. The complete dosimetry and monitoring system was successfully tested at the multidisciplinary in-air beamline of the LNS-INFN using high-intensity pulsed proton beams with dose rates ranging from 20 Gy/s to 230 Gy/s covering a wider dose rate range than previously reported. Compared to prior art (e.g., [16,24,25]), the developed system combines dosimetry with online monitoring, ensuring dose rate independence, stability, and reproducibility across the full dynamic range relevant for FLASH-RT beams. Additionally, the studies currently available in the literature generally focus on the characterization of individual detectors or on specific dosimetric configurations, often limited to a narrow dose-rate range. To our knowledge, no work has yet demonstrated the integration and cross-calibration of multiple detectors within a unified, real-time dosimetric system experimentally validated over a broad UHDR proton beam range (20–230 Gy/s).

2. Material and methods

2.1. Dosimetric system

The proposed system is specifically designed to provide measurements of the dose delivered in water at a designated *irradiation point*. This system integrates three real-time complementary detectors, each serving a specific role to ensure accurate dose monitoring throughout the irradiation process:

- The Faraday Cup serves as the primary detector delivering dose measurements directly at the *irradiation point*.
- The Secondary Emission Monitor and the Dual-Gap Ionization Chamber constitute the dosimetric monitoring system. These detectors operate continuously during irradiation, providing reliable real-time data on dose variations.

The SEM and DGIC are initially calibrated against the Faraday Cup (FC) to ensure consistency and accuracy in dose measurements. Once calibrated, they function as reliable devices for continuous monitoring of the delivered dose throughout the irradiation. This configuration enables precise real-time tracking, where the SEM and DGIC provide continuous monitoring data, while the FC serves as the reference for dose evaluation. Fig. 1 presents a possible configuration that can be implemented using this monitoring and dosimetry system. The SEM is placed in vacuum, just before the beam exits into the air, allowing it to accurately monitor its current, and therefore the dose, before the beam undergoes any changes on its way to the *irradiation point*. The proton beam is then monitored by the DGIC, an ionization chamber specifically designed for high-intensity proton beams and equipped with two air gaps to correct for any potential ion recombination effects [26]. Finally, the system includes a FC specifically designed for ultra-intense, short-duration beams, with a very low internal capacitance to maintain the highest possible signal-to-noise ratio under all operating conditions. The beam configuration in such a system is assumed to be provided by the chosen accelerator or an appropriate collimation system, as shown in Fig. 1.

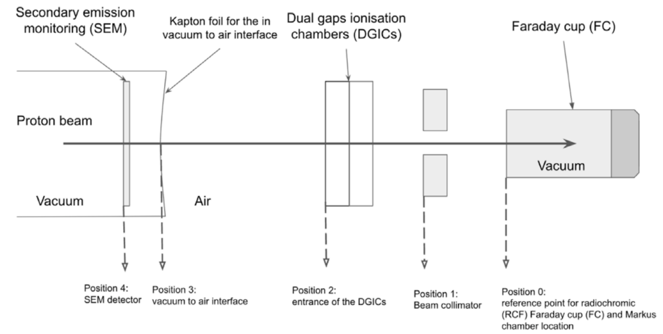

Fig. 1. Schematic representation of a potential configuration that can be implemented for a dosimetric measure of pulsed, high dose-rate proton beams with the proposed detector chain.

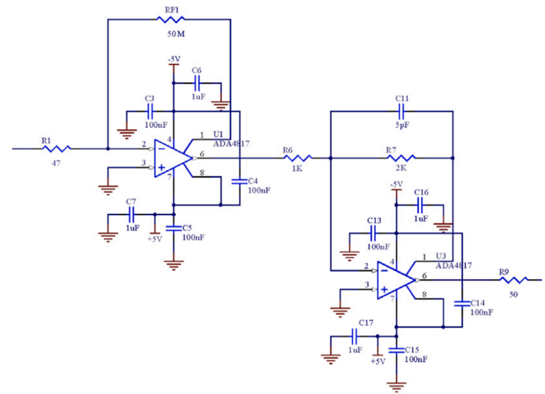

Fig. 2. Technical diagram of the trans-impedance amplifier designed for signal acquisition from both SEM and FC detectors.

Each component of the system is equipped with specialized electronic readout optimized for pulsed signals. These systems are calibrated for bunched beams lasting from a few milliseconds to several hundred milliseconds but can also be adapted to durations ranging from a few nanoseconds to several hundred nanoseconds, as well as to continuous beams. A description of the detectors composing the dosimetry and monitoring system, including the readout electronic system, is below reported.

2.2. Secondary Emission Monitor (SEM)

The Secondary Emission Monitor detector consists of a tantalum foil, 15 μm in thickness. It is electrically insulated from the beamline by a polymethylmethacrylate (PMMA) frame. The SEM works by measuring the secondary electron emission (SEE) produced when multi-MeV charged particles pass through the tantalum foil. The produced signal is proportional to the incident beam current and, with appropriate cross-calibration against a reference dosimeter, can provide real-time information directly correlated to the dose delivered at the *irradiation point*.

2.3. Dual-Gaps Ionization Chamber (DGIC)

The dual-gap ionization chamber, manufactured by DE.TEC.TOR. S.r.l. (Turin, Italy) according to the specifications provided by the authors of this paper, represents an in-transmission device designed for real-time dose monitoring over a wide range of incident proton beam currents (detector-group.com) . This device features two adjacent ionization chambers with distinct inter-electrode gaps of 5 mm (DGIC-1) and 10 mm (DGIC-2) [27]. These distinct gaps are designed to produce different charge recombination effects and, therefore, different charge collection efficiencies, allowing the chambers to act as beam monitoring detectors [28]. The anodes consist of thin layers of 5 μm copper and 2 μm nickel, deposited on a 25 μm Kapton substrate, with a total sensitive area of 240 × 240 mm^2. The cathodes consist of a 12 μm-thick layer of aluminized mylar. Each anode and cathode is mounted on rigid square fiberglass frames to ensure mechanical stability and strength. An electric field of up to ±200 V/mm can be applied via coaxial cables from a power supply unit, enabling independent polarization of the cathodes to either negative or positive voltages [29,30]. The presence of two distinct inter-electrode gaps enables the assessment of ion recombination at high dose rates by directly applying the Boag–Wilson theory to determine the charge collection efficiency f [31,32]. This approach is particularly effective under conditions where ion recombination occurs due to high radiation fluxes and is expressed through the following equation :

$$f = \frac{C}{C_{sat}} = \frac{2}{1 + \sqrt{1 + \frac{2}{3}\zeta^2}} \tag{1}$$

with

$$\zeta = m\frac{d^2\sqrt{q}}{V} \tag{2}$$

In Eq. (1), C represents the charge collected in the chamber during a given time interval as the radiation passes through; C_{sat} is the charge that would be measured in the absence of recombination; d is the distance between the electrodes; V is the applied voltage; q is the volumetric ionization rate density; and m is a constant.

2.4. Faraday cup (FC)

The Faraday Cup incorporates innovative geometrical solutions to enhance charge collection efficiency and to minimize measurement uncertainties [27]. It features an internal capacitance of 33 pF and two coaxial electrodes, with the internal electrode set to −800 V and the external electrode to +1500 V. The internal electrode, designed with a beveled geometry, contributes to creating a strong transverse electric field component, which optimizes charge collection and improves overall detector performance [33,34]. This configuration optimizes the deflection of the secondary electrons produced at both the thin entrance window and at the final collecting cup. The integrated current Q (expressed in Coulomb) collected by the FC, is converted into dose value in water (D_w, in Gy), according to Eq. (3):

$$D_w = \frac{S(E)_w}{A}\frac{Q}{e}1.602 * 10^{-10} \tag{3}$$

where A (cm^2) is the effective beam area, $S(E)_w$, expressed in MeV cm^2 g^{-1}, is the mean mass stopping power of the proton in water at the energy the proton has at the irradiation-point, and e is the elementary charge. The error on the dose is assessed by propagating the errors through Eq. (3) and taking into account the relative errors associated with the measured charge, the beam spot size at the FC entrance window, and the stopping power. The Faraday Cup is used as a reference detector for SEM and DGIC calibration in terms of dose in water at the irradiation point.

2.5. Electronic readout systems

A multipurpose trans-impedance amplifier (TA), whose electronic scheme is shown in Fig. 2, has been designed and realized at LNS-INFN to amplify the FC and SEM signals. The TA is based on two voltage feedback operational amplifiers with FET inputs, wide bandwidth and low noise performance (ADA4817, Analog Devices Inc., Norwood, MA, USA).

The TA is powered by a dual ±5 V supply and it consisted essentially of two interconnected blocks:

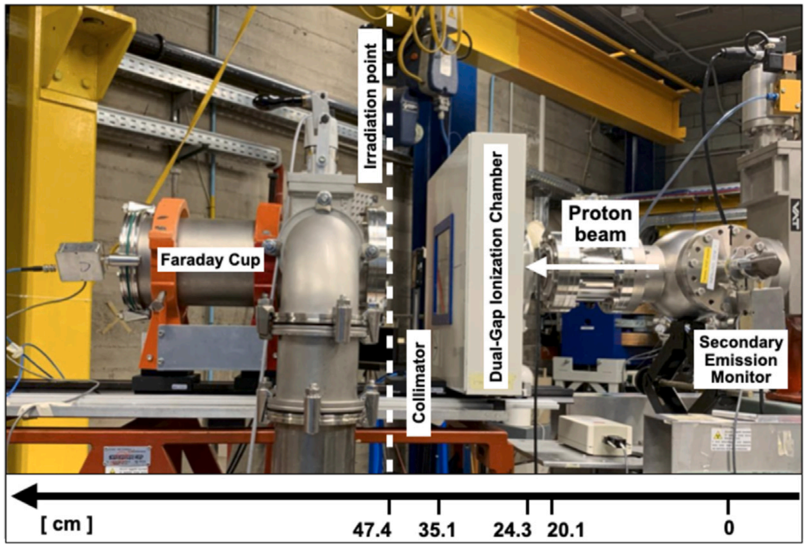

Fig. 3. Picture of the experimental setup implemented at the multidisciplinary in-air beamline of LNS-INFN. The proton beam enters from the right side of the beamline, passes through the SEM detector, and exits into the air. Immediately downstream of the beam's exit point, the setup includes the DGIC, a rectangular collimator, and the FC. The designated irradiation point is marked by the white vertical dashed line, aligned with the entrance window of the FC detector.

1. an I/V converter with an input impedance of 50 Ω, trans-impedance gain of 50 MΩ, frequency bandwidth from DC to 8 kHz (−3db), Rise/Fall time response of 45 μsec (TR from 10% to 90%, TF vice versa), output voltage noise < 1.5 mV peak-to-peak in the bandwidth from DC to 100 kHz;
2. a voltage amplifier with gain 2, capable of driving a 50 Ω transmission line.

The bandwidth of the I/V converter is limited to 8 kHz by a 0.4 pF parasitic capacitance in the feedback loop. This capacitance ensures a stable loop regulation, preventing overshoot, ringing, or oscillations. The TA provides a total trans-impedance gain ($V_{OUT/IN}$) of 100 MΩ, with a linear operating output voltage range of ±3.8 V, corresponding to an input current range of ±38 nA. No dedicated amplifier was used for the DGIC signals; the output currents of the DGIC are two orders of magnitude higher than those from the FC and SEM, and were thus directly measured with a fast oscilloscope.

2.6. Irradiation procedure

All measurements and results reported in this work were performed using a monochromatic 62 MeV proton beam accelerated by the superconducting cyclotron at INFN-LNS in Catania, Italy, and transported along the multidisciplinary in-air beamline. To produce pulsed proton beams with durations ranging from 10 ms to 200 ms, a low-energy chopper was installed along the injection line before the beam entered the cyclotron [35]. The maximum current of each proton pulse was controlled by adjusting the beam intensity at the cyclotron extraction point, spanning a range of 5 nA to 50 nA. This variation corresponded to an instantaneous dose rate in water at the irradiation-point ranging between 20 Gy/s and 230 Gy/s. The specific characteristics of the beam, measured at the irradiation-point, are detailed in Table 1.

The experimental setup, including all detector positions, is illustrated in Fig. 3 and follows the schematic shown in Fig. 1. The proton beam was first intercepted by the SEM in vacuum and then exited into air through a 50 μm-thick Kapton window, serving as the vacuum–air interface. Immediately downstream of the air interface, the beam entered the DGIC, mounted in air to monitor the beam's properties under realistic operating conditions. Following the DGIC, a 10×10 mm^2 brass aperture collimated the beam before it reached the irradiation point, where the Faraday Cup was precisely positioned to provide the reference dose calibration for the entire monitoring chain. A 30 cm air gap separates the Kapton window and the DGIC due to the fixed beamline geometry and mechanical support constraints. Although this distance is relatively large, it was rigorously characterized: beam profiles were measured at several points to confirm that no significant broadening or degradation, caused by air scattering or divergence, occurred during transit.

Table 1
Characteristics of the proton beam used for system calibration and characterization.

Instantaneous current [nA]	5–50
Pulse duration [ms]	10–200
Instantaneous dose rate [Gy/s]	20–230
Beam spot shape and size [mm^2]	Squared, 9.8 x 9.8
Beam homogeneity [%] at the irradiation point	17 (horizontal), 13 (vertical)
Nominal energy [MeV]	60 ± 0.5

The instantaneous dose rate was defined as the dose in water delivered per unit time at the irradiation point during each pulse. This was calculated from the measured charge using the Faraday Cup, the known pulse duration, and the effective beam area (as detailed in Table 1). Beam homogeneity was assessed using EBT3 radiochromic films, revealing dose uniformity values of 17% (horizontal) and 13% (vertical) across the irradiation field. These values were considered in the interpretation of all dosimetric measurements, which refer to the central beam axis. The irradiation procedure consists of two steps: first, the full dosimetry and monitoring system was calibrated, and then the independence of each detector's response from the dose rate was verified by performing irradiations with pulsed and high-intensity beams. Radiochromic films were used exclusively for preliminary evaluations of beam profile and detector positioning. They were not employed for calibration of reference dose measurements, which were solely based on the Faraday Cup and cross-calibration with the online detectors.

3. Results

Before undertaking the calibration of the full dosimetry and monitoring system, the accuracy of the dose measured by the Faraday Cup at the irradiation-point was verified. EBT3-type radiochromic films, which are notably dose rate independent detector [10], were used as the reference dosimeter, having been previously calibrated according to a well-established procedure [36]. Specifically, radiochromic films

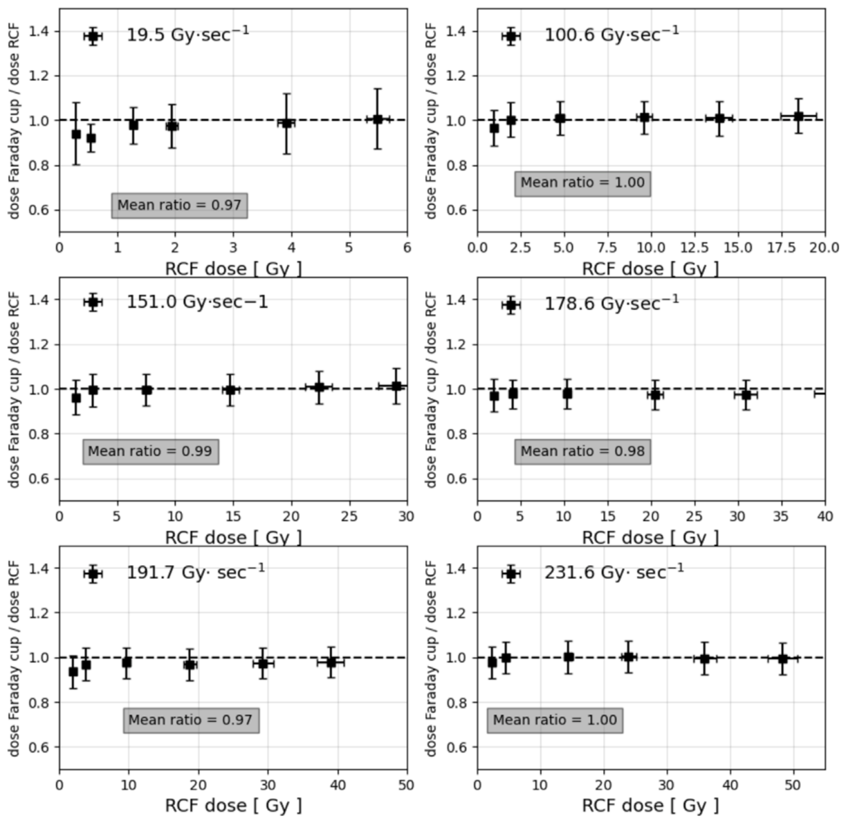

Fig. 4. Ratio of the reference doses measured by the FC to those measured with the RCFs is plotted as a function of the delivered dose. Each subplot represents a different proton dose-rate, allowing for a comparative analysis across varying irradiation conditions. The average ratio for each case is indicated by the gray rectangles within each plot, providing a visual reference for the overall consistency and potential deviations in the measurements.

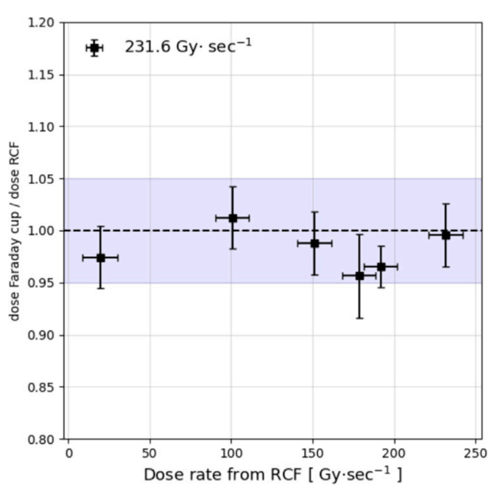

Fig. 5. Experimental ratio of the doses measured with the Faraday Cup (FC) and the Radiochromic Films (RCFs) as a function of the proton beam dose rate. The colored band represents a ±5% variation relative to the unitary ratio between the FC and RCF signals. A ratio of one indicates perfect agreement between the two detectors in measuring the same dose value. The colored band highlights that the maximum observed variation remains within 5%, demonstrating the consistency of the measurements.

were placed on the entrance window of the FC, corresponding to the irradiation-point, as shown in Fig. 3. It was assumed that the energy losses of the 60 MeV protons passing through the thin RCF plastic material were negligible. The dose measurements obtained from both detectors were subsequently compared under identical irradiation conditions. The results are presented in Fig. 4, showing the ratio between the reference doses measured by the FC and the RCFs, plotted against the reference dose from the RCFs. Each subplot corresponds to a different proton dose rate, ranging from 19.5 Gy/s to 231.6 Gy/s. The average value of the measured ratio is indicated within the gray rectangles in each plot.

The doses estimated by the FC and RCFs were in agreement within the experimental uncertainties, with a maximum discrepancy of 7.84% at the lowest dose rate (19.5 Gy/s) and an overall mean difference of 2.09% when averaging all data points. A good linearity (correlation coefficient $R^2 \geq 0.99$) as a function of the dose delivered over the range of 3 Gy to 50 Gy was accordingly observed. The ratio of the FC response to that of the RCFs was also evaluated as a function of dose rate (see Fig. 5).

The dose ratio, as reported in Fig. 5, remained approximately 1, with the largest relative difference (4.36%) observed at a dose rate of 178.6 Gy/s, and an overall mean relative difference of 2.21%. No significant trend was observed with increasing dose rates, indicating that beam intensity does not significantly affect the FC response, up to the highest dose rate explored (231.6 Gy/s). The short-term reproducibility of the FC, in terms of the integrated charge, was evaluated by performing ten consecutive irradiations under identical dose rate and pulse duration conditions, yielding a variation of less than 2.5%, demonstrating the robustness of the applied procedure. However, this comparison was only qualitative and intended for setup validation; all quantitative calibration and uncertainty analysis were based exclusively on the Faraday Cup and online detectors. After confirming the proper operation of the Faraday Cup and its independence from dose rate variations, the SEM was calibrated and its dosimetric performance validated. A series of irradiations were conducted, simultaneously recording the signals from both the SEM and the FC while varying the total dose delivered at a fixed dose rate. This allowed for the determination

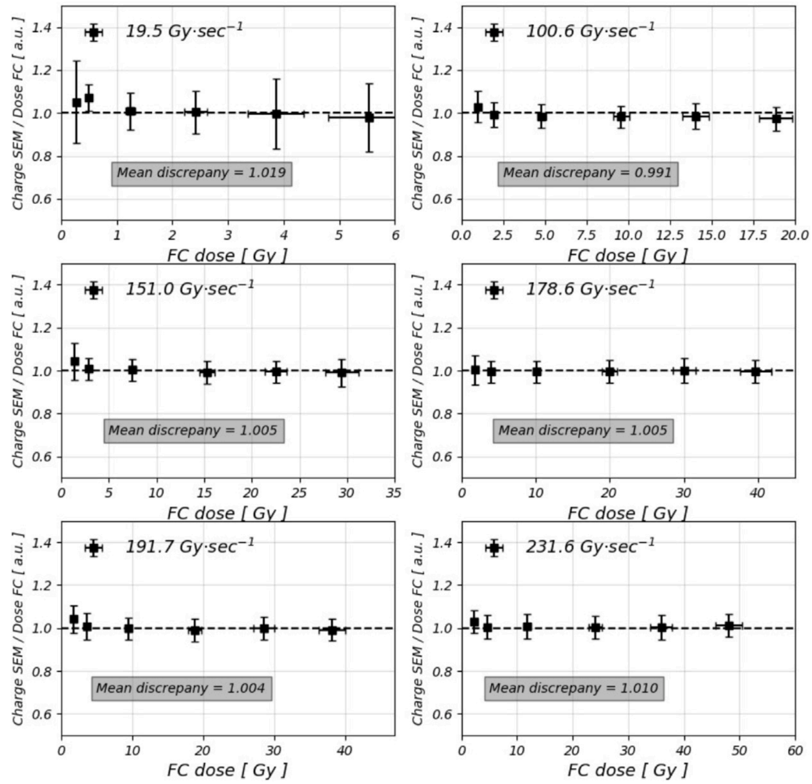

Fig. 6. Ratio of the charges measured by the SEM to those measured by the FC is presented as a function of the delivered dose. Each subplot corresponds to a distinct proton dose rate, enabling a detailed comparison of the detector responses under varying irradiation conditions. The gray rectangles in each plot indicate the average ratio, providing a reference for assessing measurement consistency and potential deviations across different dose rate scenarios.

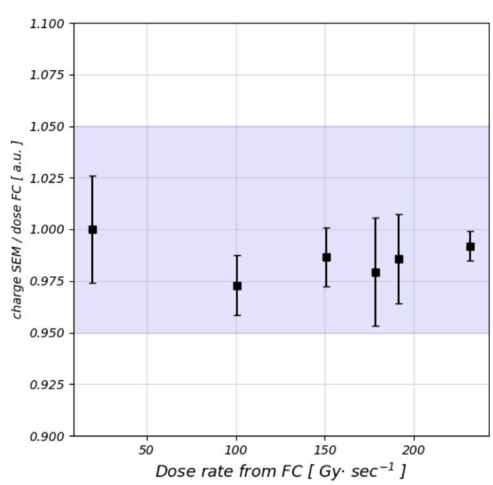

Fig. 7. Experimental ratio of the charge collected by the SEM to the dose measured by the FC as a function of the beam dose rate. The solid line denotes the mean ratio, while the dashed lines ($\pm 1\sigma$) represent the standard error of the mean across the six measurements.

of a calibration factor (Gy/C) which enable to convert the collected charge into absorbed dose in water. The resulting calibration factor was: $3.4053 \pm 0.0067 \times 10^{10}$Gy/C. With the appropriate calibration factor established, the SEM's response to varying both the total released dose and dose rates was examined. Fig. 6 reports the ratio between the SEM signal and the FC dose measure as a function of the total released dose for six different dose rates of the incident pulsed proton beam. The mean value of the observed ratio is depicted inside the gray rectangles in each plot.

The experimental results demonstrate good agreement between the doses measured by the SEM and FC across all examined beam intensities. The two detectors exhibited an average agreement within 1.44%, with a maximum relative difference of 7.27% observed at a dose rate of 19.5 Gy/s. However, the relative difference never exceeds 2.84% for doses exceeding 1.8 Gy. Additionally, the sensitivity of the SEM, defined as the detector's response per unit of released dose in water at the irradiation point, was determined by averaging across the six available dose rates. The estimated sensitivity was 0.0294 nC/cGy, with a coefficient of variation of less than 1%. With the sensitivity established, the SEM's dependence on dose rate was further investigated using the FC as a reference (see Fig. 7). No significant dependence on the investigated dose rate range was observed, suggesting that the SEM is a reliable detector for dosimetric monitoring in very high-intensity proton beams.

Finally, the calibration and testing of the entire dosimetry and monitoring system were completed by utilizing the DGIC. This step was crucial to ensuring the accuracy and reliability of dose measurements under experimental conditions. As previously described, this detector is equipped with two distinct gaps, enabling the correction of signals for ion recombination effects caused by high dose rates, once the charge collection efficiency is known. Consequently, the collection efficiencies f_1 and f_2 of the two chambers were firstly estimated by appropriately applying the Boag–Wilson theory (see Eq. (1)). This involved first the analyzing of the collected charge normalized to the inter-electrode gap as a function of the V/d^2 ratio across all investigated dose rates (see Fig. 8-a). Experimental data were then fitted to the Boag–Wilson curve using the χ^2 minimization method. This fitting process allowed the extraction of the saturation charge (C_{sat} in Eq. (1)) as a fitting parameter, which in turn permitted the determination of the collection efficiencies for the two chambers (see Fig. 8-b).

Subsequently, the DGIC was cross-calibrated against the FC to convert the measured charge into absorbed dose in water. The calibration

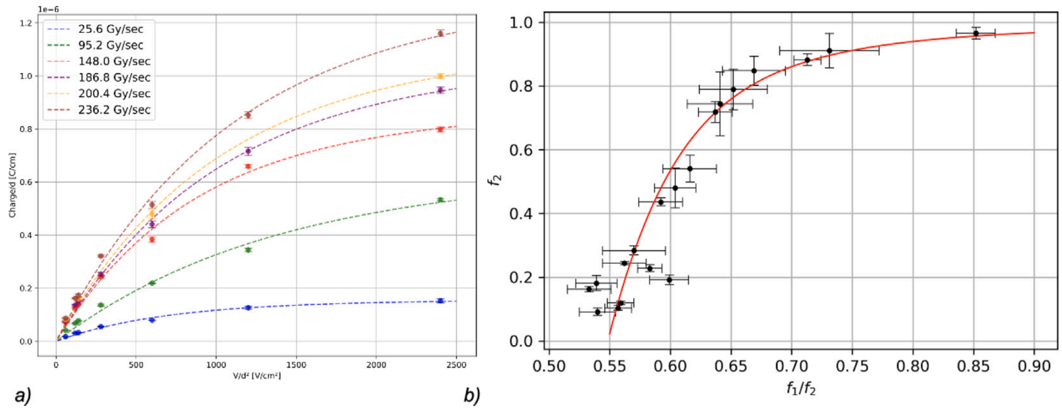

Fig. 8. (a) Collected charge, normalized to the inter-electrode gap, plotted against the V/d^2 ratio across all investigated dose rates; (b) Collection efficiency of Chamber 1 as a function of the ratio between the collection efficiencies of the two chambers.

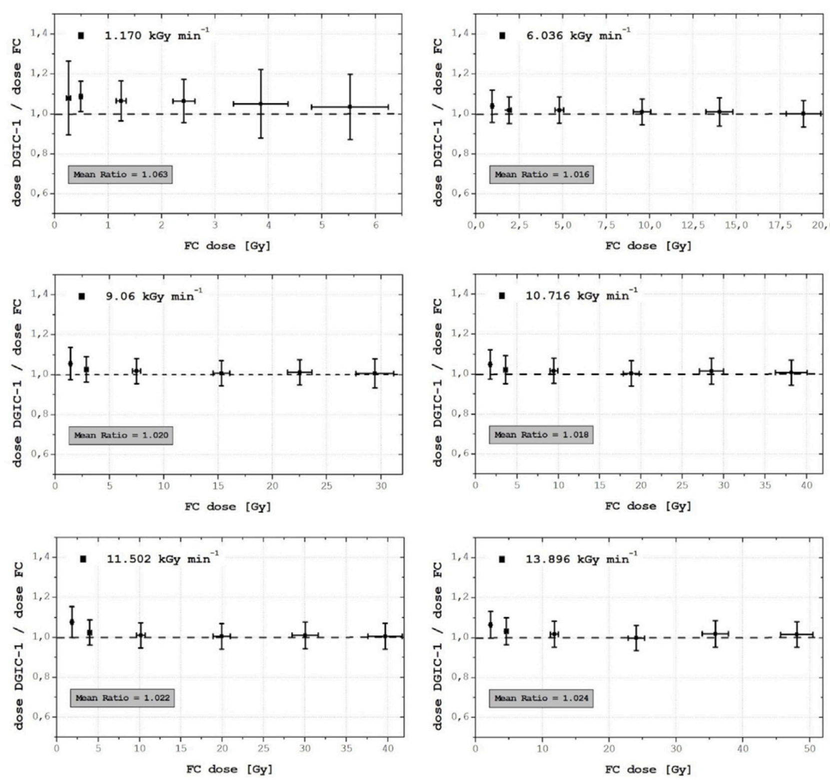

Fig. 9. Ratio of the doses measured by the DGIC-1 to those measured by the FC, plotted as a function of the released dose. Each subplot refers to a different proton dose rate. The average ratio is highlighted within the gray rectangles on each plot.

factor ($3.2490 \pm 0.1157 \times 10^7$ Gy/C) was determined by averaging the ratios of the dose measured by the FC to the charge collected by the ionization chambers at very low dose rates. At such low rates, recombination effects within the detector are considered to be negligible, with the associated uncertainty represented by the standard deviation. After converting the DGIC charge signal and determining the collection efficiency, the response's dependence on dose rate under UHDR conditions was investigated, following a similar approach to the analysis conducted for the SEM and the FC. Figs. 9 and 10 reports the ratio of the dose estimated by the DGIC-1 and the DGIC-2 ionization chambers to the dose measured by the Faraday Cup, as a function of the released dose for all investigated pulsed proton beam dose rates.

The mean deviation was measured to be less than 3%. However, a maximum variation of up to 8.65% and 8.46% was observed for DGIC-1 and DGIC-2, respectively, at the lowest dose rate of 1.170 kGy/min. The agreement between the ionization chambers and the FC improved with increasing delivered dose, reaching a discrepancy of less than 2% for doses typically greater than 5 Gy per shot. The smaller ionization chamber (DGIC-2) exhibited slightly enhanced accuracy in dose estimation, with mean discrepancies relative to the FC averaging 2.75% for DGIC-1 and 2.59% for DGIC-2 across the six pulse proton beam dose rates.

4. Discussion and conclusion

The main objective of this study was to demonstrate the feasibility of a detector chain specifically designed for beam monitoring and dosimetry in the UHDR regime of pulsed proton beams, with a view toward future clinical and radiobiological implementation. The proposed dosimetry and monitoring system combines three detectors: a Secondary Emission Monitor, a Dual-Gap Ionization Chamber, and a Faraday Cup. Experimental measurements were conducted using a 62

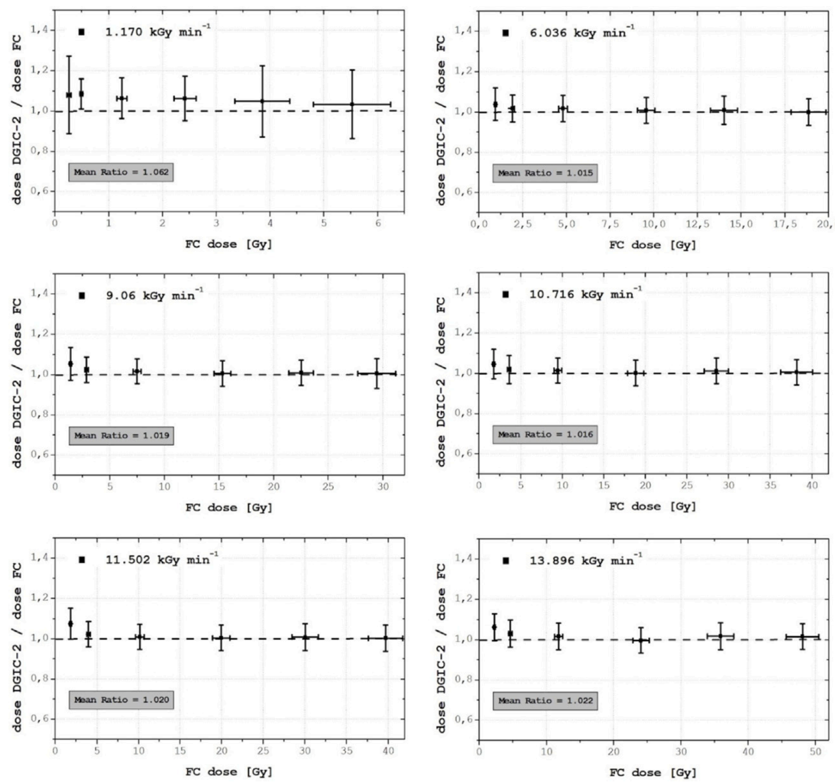

Fig. 10. Ratio of the doses measured by the DGIC-2 to those measured by the FC, plotted as a function of the released dose. Each subplot refers to a different proton dose rate. The average ratio is highlighted within the gray rectangles on each plot.

MeV pulsed proton beam with dose rates ranging from 20 Gy s^{-1} up to 231.6 Gy s^{-1}. The SEM and FC demonstrated stable response across the full intensity range tested, with no significant dependence on beam current, confirming their robustness and suitability for high-intensity beam monitoring. The DGIC, although affected by ion recombination at the highest dose-per-pulse values, achieved accurate dose measurements through the application of Boag–Wilson recombination corrections. Quantitatively, the overall accuracy of the dose estimation, expressed as the mean relative difference between the three detectors, remained within 3%. This level of agreement satisfies clinical tolerances for dosimetric systems and supports the reliability of the proposed solution for proton FLASH applications. Furthermore, the system enables real-time and redundant dose verification, which is essential in the context of UHDR delivery. However, several limitations of the present study must be acknowledged. Dose measurements with the FC were benchmarked against Gafchromic EBT3 film, which, while practical and commonly used, does not constitute a high-accuracy reference standard. Radiochromic films suffer from limited statistical power and higher uncertainty compared to primary-standard methods such as calorimetry or well-characterized ionization chambers (e.g., the IBA PPC05 or the recently developed designs by Gómez et al. [24]). The reported agreement of approximately 8% between FC and film falls within experimental uncertainty but is not sufficient to validate the FC as a high-precision reference device. This confirms that our study represents a feasibility demonstration rather than a fully validated clinical-grade solution. Moreover, the system was tested in a single energy configuration (62 MeV) and with a passively scattered beam. Its performance under other clinically relevant conditions, such as pencil beam scanning (PBS), energy modulation, and full treatment fields, remains to be evaluated. Similarly, the effect of environmental factors (e.g., temperature stability, humidity) and long-term detector reproducibility have not yet been fully characterized. Looking ahead, several developments are planned. Most notably, we intend to benchmark our diagnostic and dosimetric system against graphite calorimeter. This will provide a primary-standard validation of the FC and DGIC performance under UHDR conditions, significantly improving the system traceability. Additionally, we plan to test the chain with scanned beams and clinical treatment plans to assess its integration into real-time monitoring and quality assurance workflows. In conclusion, the proposed dosimetry and monitoring system, supported by an appropriate calibration and correction protocol, presents a feasible, modular, and real-time solution for dosimetry in proton FLASH beams. While further improvements and validations are necessary, particularly with respect to absolute accuracy and standardization, this work provides a solid foundation toward the development of clinically usable UHDR dosimetric systems.

CRediT authorship contribution statement

Giada Petringa: Participated in the experiment, Contributing to the experimental setup, Online data acquisition, Analysis, Drafting the manuscript, and revising it. **Roberto Catalano:** Contributed to the experiment setup and data acquisition, Performed data analysis, Specifically focusing on the reconstruction of DGIC signals. **Antonino Amato:** Developing the detector readout electronics, Ensuring data acquisition with a high signal-to-noise ratio throughout the experimental measurements. **Antonio Domenico Russo:** Essential support in the installation and optimal functioning of the chopper, which generated the pulsed beams at LNS. **Giuseppe Francesco Fustaino:** Experiment and participated in data acquisition. **Mariacristina Guarrera:** Experiment and participated in data acquisition. **Giacomo Cuttone:** Critical revision of the manuscript. **Gustavo Esteban Messina:** Responsible for the fabrication, Precise alignment of the mechanical components supporting the detectors. **Luigi Raffaele:** Critical revision of the manuscript. **Alfio Domenico Pappalardo:** Critical revision of the manuscript. **Giuseppe Antonio Pablo Cirrone:** Experiment by designing the conceptual framework of the measurement, Assisting with the experimental setup, Drafting and revising the manuscript, Generating the final plots included in the paper.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Bourhis J, Montay-Gruel P, Goncalves Jorge P, Bailat C, Petit B, Ollivier J, Jeanneret-Sozzi W, Ozsahin M, Bochud F, Moeckli R, Germond JF, Vozenin MC. Clinical translation of FLASH radiotherapy: Why and how? Radiother Oncol 2019;139:11–7.

[2] Durante M, Brauer-Krisch E, Hill M. Faster and safer? FLASH ultra-high dose rate in radiotherapy. Br J Radiol 2017;20170628.

[3] Tinganelli W, Sokol O, Quartieri M, Puspitasari A, Dokic I, Abdollahi A, Durante M, Haberer T, Debus J, Boscolo D, Voss B, Brons S, Schuy C, Horst F, Weber U. Ultra-high dose rate (FLASH) carbon ion irradiation: Dosimetry and first cell experiments. Int J Radiat Oncology*Biology*Physics 2022;112(4):1012–22.

[4] Tessonnier T, Mein S, Walsh DW, Schuhmacher N, Liew H, Cee R, Galonska M, Scheloske S, Schömers C, Weber U, Brons S, Debus J, Haberer T, Abdollahi A, Mairani A, Dokic I. FLASH dose rate helium ion beams: First in vitro investigations. Int J Radiat Oncology*Biology*Physics 2021;111(4):1011–22.

[5] Harrington KJ. Ultrahigh dose-rate radiotherapy: Next steps for FLASH-RT. Clin Cancer Res 2019;25(1):3–5.

[6] Bourhis J, Sozzi WJ, Jorge PG, Gaide O, Bailat C, Duclos F, Patin D, Ozsahin M, Bochud F, Germond JF, Moeckli R, Vozenin MC. Treatment of a first patient with FLASH-radiotherapy. Radiother Oncol 2019;139:18–22.

[7] Mascia AE, Daugherty EC, Zhang Y, Lee E, Xiao Z, Sertorio M, Woo J, Backus LR, McDonald JM, McCann C, Russell K, Levine L, Sharma RA, Khuntia D, Bradley JD, Simone CB, Perentesis JP, Breneman JC. Proton FLASH radiotherapy for the treatment of symptomatic bone metastases: The FAST-01 nonrandomized trial. JAMA Oncol 2023;9(1):62.

[8] Darafshes A. Radiation therapy dosimetry: a practical handbook. Boca Radon: CRC Press; 2021.

[9] IAEA. Absorbed dose determination in external beam radiotherapy. Technical reports series no. 398 (rev. 1), International Atomic Energy Agency; 2024.

[10] Romano F, Bailat C, Jorge PG, Lerch MLF, Darafsheh A. Ultra-high dose rate dosimetry: Challenges and opportunities for FLASH radiation therapy. Med Phys 2022;49(7):4912–32.

[11] Lee E, Lourenço AM, Speth J, Lee N, Subiel A, Romano F, Thomas R, Amos RA, Zhang Y, Xiao Z, Mascia A. Ultrahigh dose rate pencil beam scanning proton dosimetry using ion chambers and a calorimeter in support of first in-human FLASH clinical trial. Med Phys 2022;49(9):6171–82.

[12] Ceberg S, Mannerberg A, Konradsson E, Blomstedt M, Kugele M, Kadhim M, Edvardsson A, Back SAJ, Petersson K, Jamtheim Gustafsson C, Ceberg C. FLASH radiotherapy and the associated dosimetric challenges. J Phys: Conf Ser 2023;2630(1):012010.

[13] Siddique S, Ruda HE, Chow JCL. FLASH radiotherapy and the use of radiation dosimeters. Cancers 2023;15(15):3883.

[14] Diffenderfer ES, Verginadis II, Kim MM, Shoniyozov K, Velalopoulou A, Goia D, Putt M, Hagan S, Avery S, Teo K, Zou W, Lin A, Swisher-McClure S, Koch C, Kennedy AR, Minn A, Maity A, Busch TM, Dong L, Koumenis C, Metz J, Cengel KA. Design, implementation, and in vivo validation of a novel proton FLASH radiation therapy system. Int J Radiat Oncology*Biology*Physics 2020;106(2):440–8.

[15] Schoenauen L, Coos R, Colaux JL, Heuskin AC. Design and optimization of a dedicated faraday cup for UHDR proton dosimetry: Implementation in a UHDR irradiation station. Nucl Instrum Methods Phys Res Sect A: Accel Spectrometers Detect Assoc Equip 2024;1064:169411.

[16] Lourenço A, Subiel A, Lee N, Flynn S, Cotterill J, Shipley D, Romano F, Speth J, Lee E, Zhang Y, Xiao Z, Mascia A, Amos RA, Palmans H, Thomas R. Absolute dosimetry for FLASH proton pencil beam scanning radiotherapy. Sci Rep 2023;13(1).

[17] Pedroni E, Scheib S, Böhringer T, Coray A, Grossmann M, Lin S, Lomax A. Experimental characterization and physical modelling of the dose distribution of scanned proton pencil beams. Phys Med Biol 2005;50(3):541–61.

[18] Lin S, Boehringer T, Coray A, Grossmann M, Pedroni E. More than 10 years experience of beam monitoring with the Gantry 1 spot scanning proton therapy facility at PSI. Med Phys 2009;36(11):5331–40.

[19] Kacperek A. Clinical proton dosimetry part I: Beam production, beam delivery and measurement of absorbed dose (ICRU Report 59). Phys Med Biol 2000;45(10):3123–4.

[20] Grusell E, Isacsson U, Montelius A, Medin J. Faraday cup dosimetry in a proton therapy beam without collimation. Phys Med Biol 1995;40(11):1831–40.

[21] Verhey LJ, Koehler AM, McDonald JC, Goitein M, Ma IC, Schneider RJ, Wagner M. The determination of absorbed dose in a proton beam for purposes of charged-particle radiation therapy. Radiat Res 1979;79(1):34.

[22] Levin DS, Friedman PS, Ferretti C, Ristow N, Tecchio M, Litzenberg DW, Bashkirov V, Schulte R. A prototype scintillator real-time beam monitor for ultra-high dose rate radiotherapy. Med Phys 2024;51(4):2905–23.

[23] Kanouta E, Bruza P, Johansen JG, Kristensen L, Sørensen BS, Poulsen PR. Two-dimensional time-resolved scintillating sheet monitoring of proton pencil beam scanning FLASH mouse irradiations. Med Phys 2024;51(7):5119–29.

[24] Gómez F, Gonzalez-Castaño DM, Fernández NG, Pardo-Montero J, Schüller A, Gasparini A, Vanreusel V, Verellen D, Felici G, Kranzer R, Paz-Martín J. Development of an ultra-thin parallel plate ionization chamber for dosimetry in FLASH radiotherapy. Med Phys 2022;49(7):4705–14.

[25] Subiel A, Romano F. Recent developments in absolute dosimetry for FLASH radiotherapy. Br J Radiol 2023;96(1148).

[26] Kranzer R, Poppinga D, Weidner J, Schüller A, Hackel T, Looe HK, Poppe B. Ion collection efficiency of ionization chambers in ultra-high dose-per-pulse electron beams. Med Phys 2021;48(2):819–30.

[27] Cirrone G, Cuttone G, Romano F, Schillaci F, Scuderi V, Amato A, Candiano G, Costa M, Gallo G, Larosa G, Korn G, Leanza R, Manna R, Maggiore M, Marchese V, Margarone D, Milluzzo G, Petringa G, Tramontana A. Design and status of the ELIMED beam line for laser-driven ion beams. Appl Sci 2015;5(3):427–45.

[28] Giordanengo S, Guarachi LF, Braccini S, Cirrone GAP, Donetti M, Fausti F, Mas Milian F, Romano F, Vignati A, Monaco V, Cirio R, Sacchi R. Fluence beam monitor for high-intensity particle beams based on a multi-gap ionization chamber and a method for ion recombination correction. Appl Sci 2022;12(23):12160.

[29] Cirrone GAP, Petringa G, Catalano R, Schillaci F, Allegra L, Amato A, Avolio R, Costa M, Cuttone G, Fajstavr A, Gallo G, Giuffrida L, Guarrera M, Korn G, Larosa G, Leanza R, Lo Vecchio E, Messina G, Milluzzo G, Olsovcova V, Pulvirenti S, Pipek J, Romano F, Rizzo D, Russo AD, Salamone S, Scuderi V, Velyhan A, Vinciguerra S, Zakova M, Zappalà E, Margarone D. ELIMED-ELIMAIA: The first open user irradiation beamline for laser-plasma-accelerated ion beams. Front Phys 2020;8.

[30] Margarone D, Cirrone G, Cuttone G, Amico A, Andò L, Borghesi M, Bulanov S, Bulanov S, Chatain D, Fajstavr A, Giuffrida L, Grepl F, Kar S, Krasa J, Kramer D, Larosa G, Leanza R, Levato T, Maggiore M, Manti L, Milluzzo G, Odlozilik B, Olsovcova V, Perin JP, Pipek J, Psikal J, Petringa G, Ridky J, Romano F, Rus B, Russo A, Schillaci F, Scuderi V, Velyhan A, Versaci R, Wiste T, Zakova M, Korn G. ELIMAIA: A laser-driven ion accelerator for multidisciplinary applications. Quantum Beam Sci 2018;2(2):8.

[31] Boag JW. Ionization measurements at very high intensities—Part I. Br J Radiol 1950;23(274):601–11.

[32] Boag JW, Wilson T. The saturation curve at high ionization intensity. Br J Appl Phys 1952;3(7):222–9.

[33] Scuderi V, Amato A, Amico AG, Borghesi M, Cirrone GAP, Cuttone G, Fajstavr A, Giuffrida L, Grepl F, Korn G, Larosa G, Leanza R, Margarone D, Milluzzo G, Petringa G, Pipek J, Russo A, Schillaci F, Velyhan A, Romano F. Diagnostics and dosimetry solutions for multidisciplinary applications at the ELIMAIA beamline. Appl Sci 2018;8(9):1415.

[34] Romano F, Cirrone GAP, Cuttone G, Schillaci F, Scuderi V, Amico A, Candiano G, Giordanengo S, Guarachi LF, Korn G, Larosa G, Leanza R, Manna R, Marchese V, Marchetto F, Margarone D, Milluzzo G, Petringa G, Pipek J, Sacchi R, Vignati A. Status of the ELIMED multidisciplinary and medical beam-line at ELI-beamlines. J Phys: Conf Ser 2017;777:012016.

[35] Calabretta L. The radio frequency pulsing system at INFN-LNS. In: AIP conference proceedings, vol. 600, AIP; 2001, p. 297–9.

[36] Cirrone G, Petringa G, Cagni B, Cuttone G, Fustaino G, Guarrera M, Khanna R, Catalano R. Use of radiochromic films for the absolute dose evaluation in high dose-rate proton beams. J Instrum 2020;15(04). C04029–C04029.